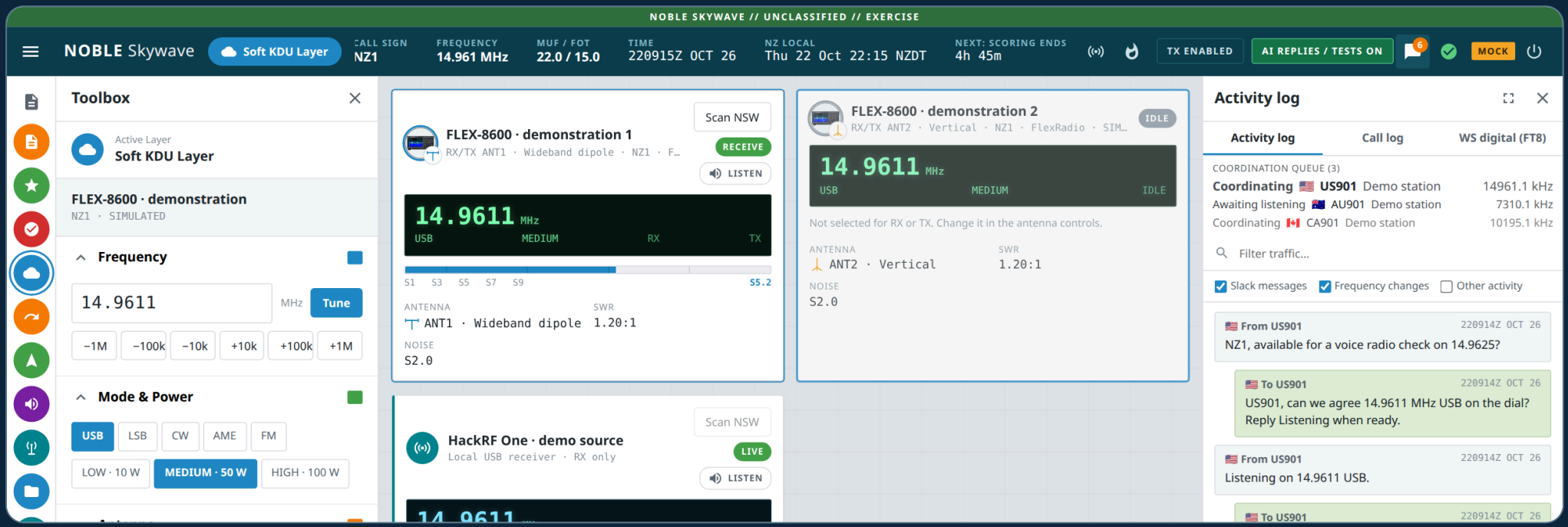
From first request to recorded contact.

A visual tour of station coordination, map-based GRAFEX frequency planning, guarded radio control, receive audio and reporting.



COORDINATE | MAP & GRAFEX | TUNE & VERIFY | TRANSMIT / RECEIVE | RECORD & REPORT

SIMULATED WALKTHROUGH

One contact. One connected workflow.

The app carries the same station, dial and evidence through every stage of the exchange.

01 Discover — Find a fresh request in main chat or a peer DM. →

02 Coordinate — Use location, GRAFEX and the plan to agree the dial and who calls first. →

03 Verify — Check the plan, actual radio, antenna, tuner and clear channel. →

04 Exchange — Make one guarded call, then open the receive window. →

05 Capture — Keep original audio, timing, source and receive settings. →

06 Confirm — Assess the two directions using the recording and peer report. →

07 Report — Prepare the contact draft and retain the delivery outcome.

Inputs

Station roster, main-chat requests, peer replies, approved channels, radio telemetry and receive audio.

Decisions

Which station is actionable, which dial is agreed, whether RX or TX is permitted, and what the evidence supports.

Outputs

Coordination messages, guarded RF checks, recordings, separate reports, local contact logs and reporting drafts.

The thread running through the app

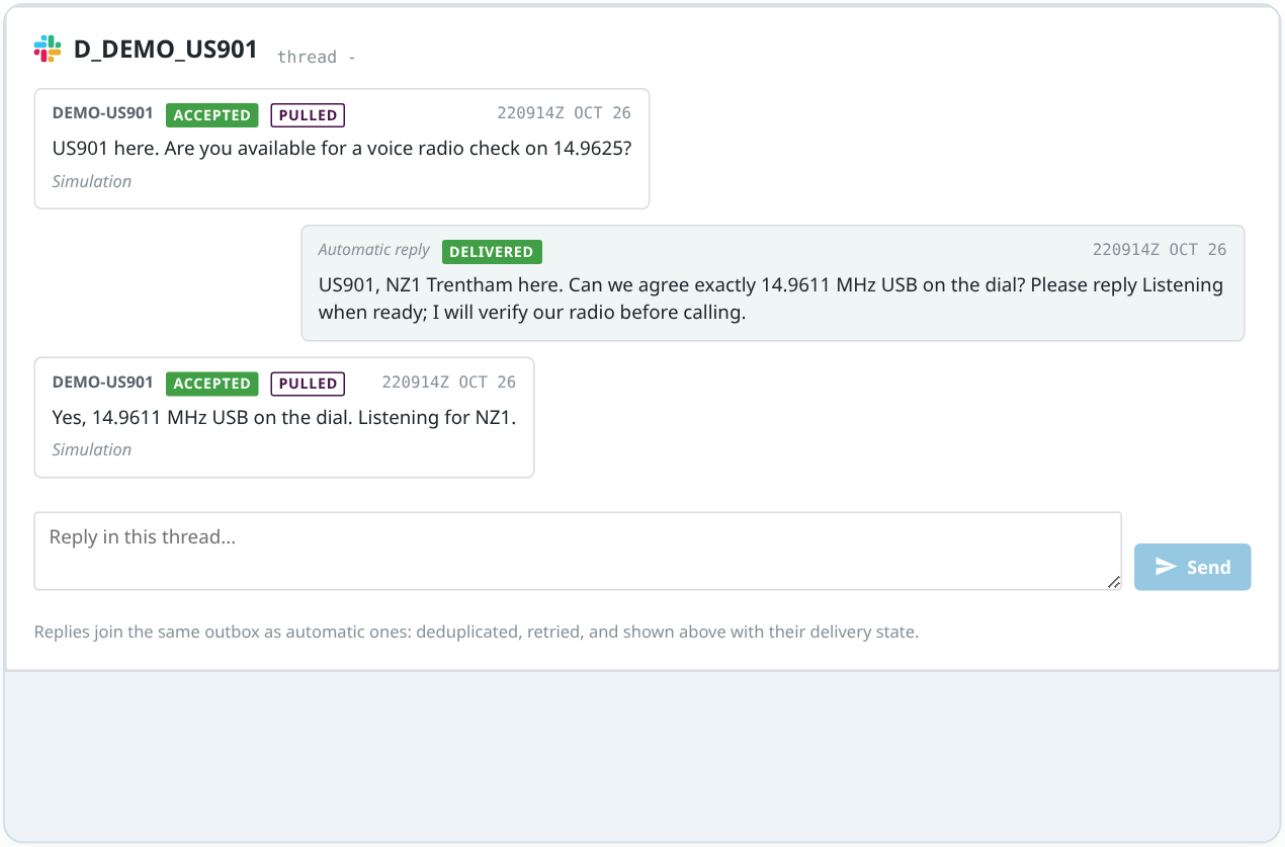
Know who is being worked, verify where the receiver really is, and keep evidence for what happened. A queue position or a sent invitation does not mean a radio call has occurred.

Demo contact: fictional US901 on 14.9611 MHz USB. Calls, RF measurements, transcripts and contact outcomes are simulated. Pages 6–10 instead use the verified Bureau NZ1→AU8 forecast and supplied plan-channel example, replayed offline.

SIMULATED WALKTHROUGH

01 / Find a station and agree the check.

A main-chat opportunity becomes a specific conversation with an identified station.



Actual Slack conversation view, populated offline with fictional US901 messages. Outgoing replies show their delivery state.

DISCOVERY → COORDINATION

Work fresh replies first.

1. **Match the identity**
   Compare the requester with the station roster before opening or continuing a peer exchange.

2. **Name the exact dial**
   Write the USB dial clearly in MHz and kHz. Confirm whether NZ1 should transmit or receive first.

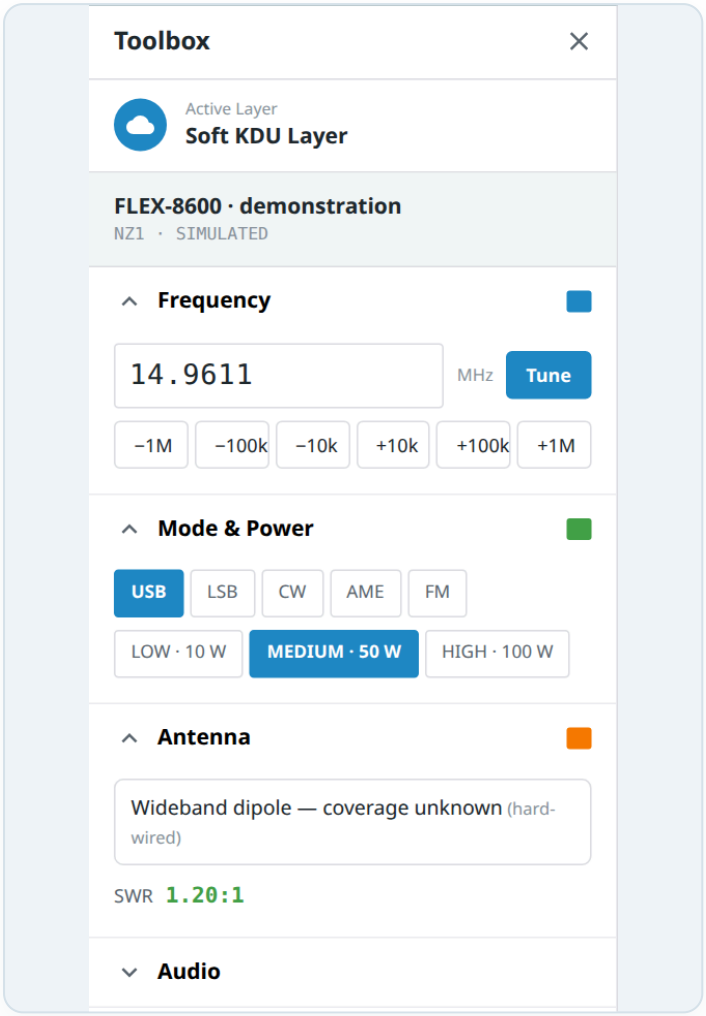
3. **Wait for readiness**
   A fresh “Listening” opens the path toward a guarded NZ1 call. If the peer is transmitting first, prepare the receiver instead.

> Several DMs can be coordinated at once. The RF exchange is serialized so stations do not compete for the same radio.

SIMULATED WALKTHROUGH

02 / Tune the radio that will do the check.

The Soft KDU connects a coordinated dial with the radio’s actual reported state.



The real Soft KDU layout: frequency, mode, power, antenna and on-air controls.

EXACT DIAL / ACTUAL READBACK

14.9611 MHz USB

Demonstration dial: 14961.1 kHz. A published channel centre and its voice dial can differ; the agreed dial must be explicit.

1. **Choose frequency and mode**
   Tune the selected source and read back the frequency and USB mode before announcing readiness.

2. **Check the selected signal path**
   Verify the receive antenna, transmit antenna and power setting. Where paired FLEX receivers are enabled, verify both are aligned.

3. **Announce what is true**
   “Listening” means the correct receiver is tuned and the receive window is actually running. A software command alone is not that proof.

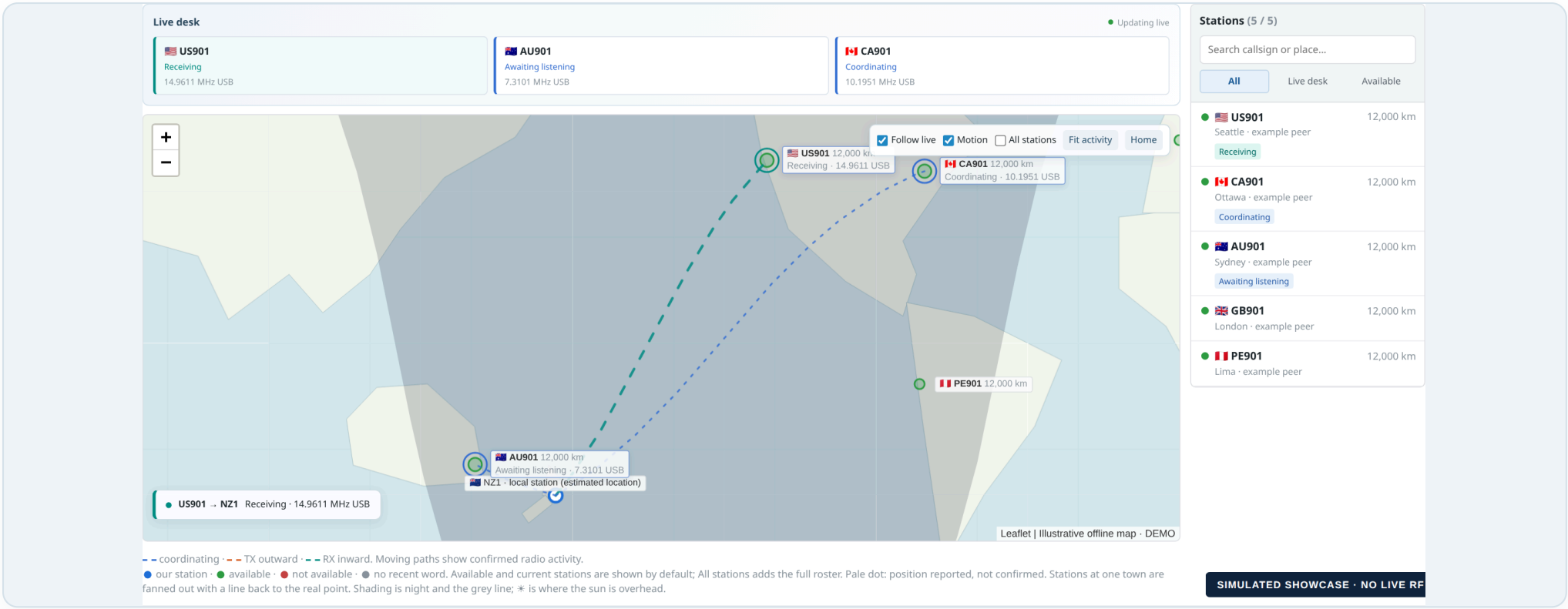
Ownership follows the radio

A live claim or manual takeover holds competing automation. The operator can inspect the same controls used by the automated desk.

SIMULATED WALKTHROUGH

03 / Use the map to see the working paths.

Live coordination and RF activity sit beside station locations and propagation planning.



Actual circuit-map UI with fictional stations and an illustrative offline basemap. The exported PDF freezes animations into one frame.

Blue / coordination

A station is being coordinated or is awaiting readiness.

Orange / transmitting

A current TX state identifies the station actually being called.

Teal / receiving

A current receive window identifies the station being monitored.

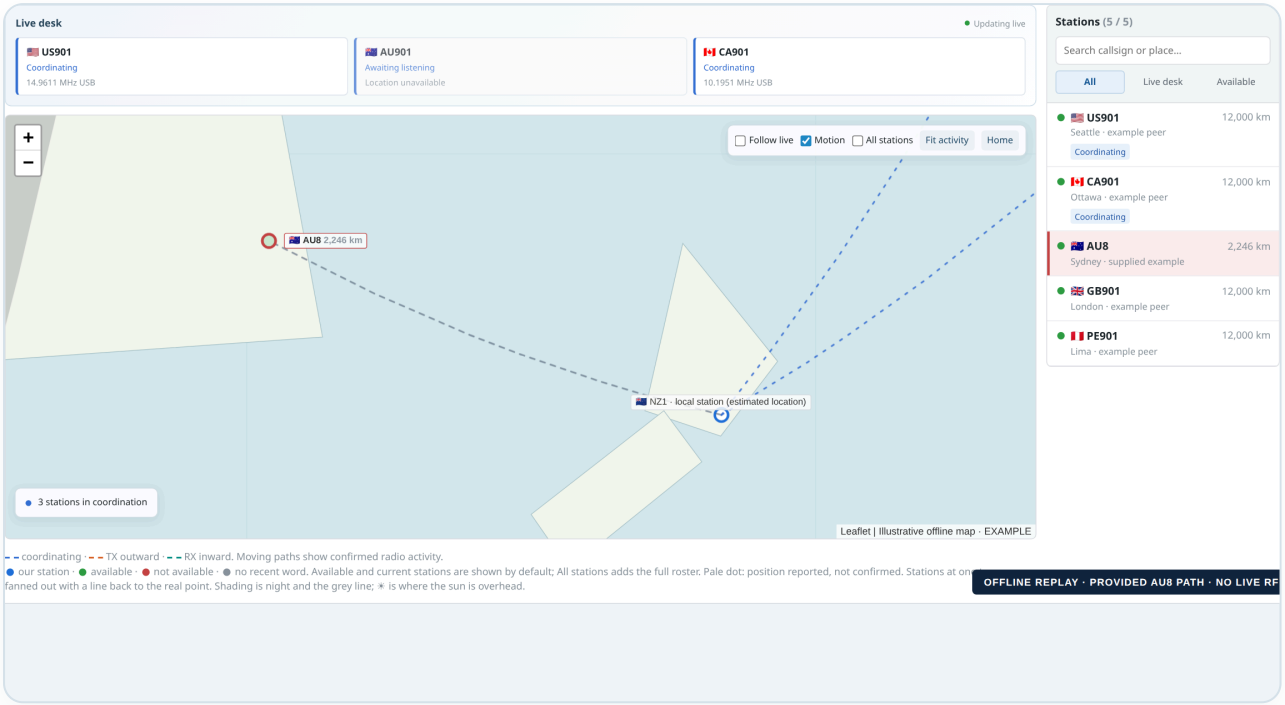
GRAFEX / planning

Select a station or a point for its hourly forecast and ranked plan channels. See the next five pages.

SIMULATED WALKTHROUGH

03A / Click a station. See its path forecast.

The map turns a selected station’s position into a location-to-location GRAFEX request.



Actual app UI, replayed offline for the supplied AU8 / Sydney path. The map base is illustrative; no live RF or peer exchange occurs.

MAP → COORDINATES → FORECAST

Choose who you want to reach.

1. **Select a station or a point**
   Click a station marker, choose a station in the list, or click another point on the map. The destination coordinates fill automatically.

2. **Ask for that specific path**
   The app sends the destination, UTC prediction date and optional T-index to its path service. The service adds NZ1’s configured location.

3. **Read the result together**
   The path summary, recommended plan channels, 24-hour chart and expandable original report appear in the same panel.

A station needs a usable position for its path to be predicted. Selecting another station requests that new circuit.

TX / NZ1

`−41.14°, 175.04°`

Configured local station position

RX / AU8, Sydney

`−33.87°, 151.18°`

Selected destination position

Forecast inputs

22 October 2026 · T-index 60

Monthly Bureau input; example view at 20Z

The Live desk strip focuses a station; map markers and station-list selections fetch forecasts. The map and forecast controls do not key the transmitter.

SIMULATED WALKTHROUGH

03B / How the app gets and processes GRAFEX.

The Bureau supplies the HF propagation prediction. The app validates it, measures the path and interprets its hourly readings.

1 · Browser
Selected station / point
UTC date + optional T-index

→

2 · App service
POST /api/propagation/path
Add configured NZ1 coordinates

→

3 · Bureau SWS
Monthly T-index lookup
GRAFEX circuit request

→

4 · App result
Validated raw report + 24 rows
Chart, path and plan ranking

THE EXTERNAL DATA REQUEST

Coordinates in; original report back.

The adapter calls the Bureau's `grafexj-cgi.tcl` endpoint with TX/RX latitude and longitude, day, month, year and T-index. Coordinates are sent to two decimal places; longitude is normalized to 0–360°.

If T-index is blank, it first reads the Bureau's monthly index data. October 2026 returns **60**; an entered value is recorded as an operator override.

Validate before drawing

Extract the HTML <pre> report; check the echoed date and coordinates, all 24 UTC rows, fixed-width layout and recognized symbols. Preserve the raw text, source URL, fetch time and parser version.

A timeout or changed report format returns an error. The chart is not filled with invented forecast data.

THE LOCATION MATH

Measure the great-circle path.

$$h = \sin^2(\Delta\varphi/2) + \cos\varphi_1 \cos\varphi_2 \sin^2(\Delta\lambda/2)$$
$$d = 2 \times 6371 \times \operatorname{asin}(\sqrt{h})$$

Latitude φ and longitude λ are converted to radians; distance d is in kilometres. The initial bearing uses atan2 and is normalized to 0–360°.

NZ1 → AU8, using the supplied coordinates

Local calculation: 2245.638 km → **2,246 km**; initial bearing 283.375° → **283°**.

Bureau report: 2,245 km, bearings **283° / 118°**. The small distance difference is retained rather than changing either source.

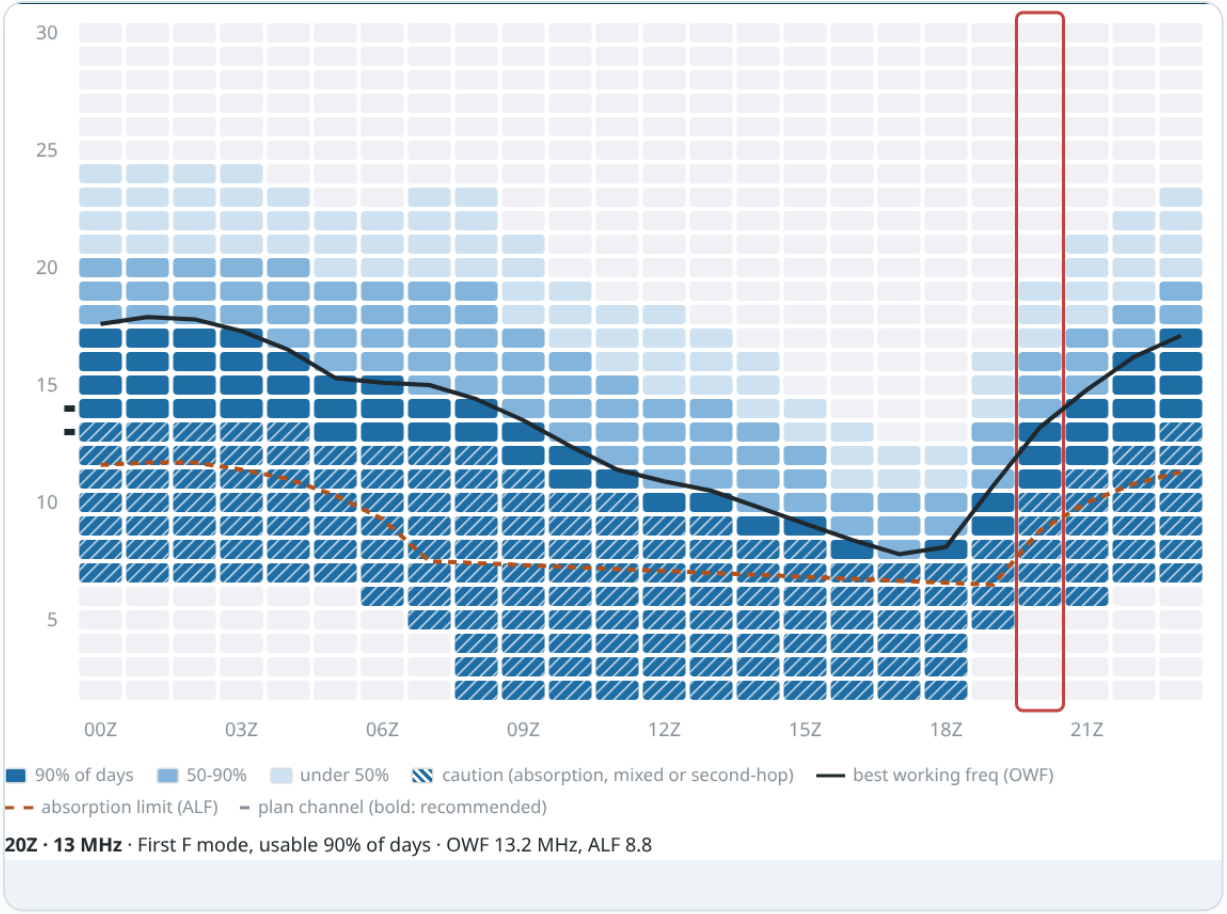
The geometry describes the circuit. The ionospheric frequency prediction comes from GRAFEX; distance alone does not determine the best frequency. If NZ1 has no configured position, the service labels its Wellington reference fallback.

Public sources: Bureau prediction tools · monthly T-index data · exact AU8 worked-example request. Implementation: `api/routes.py · propagation/grafex.py · propagation/path.py`.

SIMULATED WALKTHROUGH

03C / Turn the text into a readable 24-hour chart.

The heatmap and channel table read the same parsed Bureau report, so each visual choice can be traced back to a printed hourly symbol.



Actual app chart from the verified example report. This offline replay highlights 20Z and shows the interactive cell readout.

UTC HOUR × FREQUENCY

1 Read time and frequency

Columns run 00–23Z. The displayed vertical axis spans 2–30 MHz. The outlined column marks the current UTC hour when viewing today's date.

2 See availability and caution

Darker cells represent higher predicted availability. F means first F mode; % means usable 50–90% of days; dots mean under 50%. Hatching flags absorption, mixed, second-hop or complex modes.

3 Follow the curves and plan marks

The solid line is OWF, the optimum working frequency. The dashed line is ALF, the absorption limiting frequency. Side ticks mark plan-channel bins; bold ticks mark ranked candidates.

At 20Z in this example

First-mode **OWF 13.2 MHz**; **ALF 8.8 MHz**. Hovering a cell shows its exact printed symbol and meaning.

The source has one character per 1 MHz bin, across 1–40 MHz. There is no fine-frequency interpolation or per-call success probability. A printed ALF of 0.0 means no limit was printed for that hour.

The summary's "6–16 MHz" at 20Z is the outer span of nonblank, non-dot bins. It can include cautioned bins or gaps; inspect the cells and channel readings before choosing a dial.

SIMULATED WALKTHROUGH

03D / Rank the plan. Agree the exact USB dial.

Propagation advice narrows the choices; the operating plan and the peer’s agreement determine which frequency can actually be used.

Channel	Centre MHz	Dial (USB)	Chart, hour by hour
A2-057 best	13.4165	13.4151	20Z F, 21Z F, 22Z F, 23Z X, 00Z X, 01Z X
A2-058	13.5685	13.5671	20Z %, 21Z F, 22Z F, 23Z F, 00Z F, 01Z F
A2-059	13.5855	13.5841	20Z %, 21Z F, 22Z F, 23Z F, 00Z F, 01Z F
A2-095	13.5855	13.5841	20Z %, 21Z F, 22Z F, 23Z F, 00Z F, 01Z F
A2-060	13.8815	13.8801	20Z %, 21Z F, 22Z F, 23Z F, 00Z F, 01Z F

Actual ranking table, replayed with the five plan rows in the supplied example. The six-hour window wraps from 20Z through 01Z.

Forecast ranking → permission check → exact USB dial → peer agreement → verified radio

1 / READ & RANK

Prefer the near-term path.

Read each voice channel’s **centre** from the nearest 1 MHz chart bin. Exclude channels closed during the window. Score high / medium / low / blank as 3 / 2 / 1 / 0, subtracting one for caution.

Compare the first three hours’ minimum, then the weighted total $\Sigma\ score_i / (1 + 0.25i)$, $i = 0…5$. Earlier hours carry more weight.

2 / CHECK PERMISSION

Keep the allowed choices.

Recheck the active plan, current phase and scope, station/direction permissions, time, mode, equipment and power limits. The map’s “best” badge alone does not perform all these operational checks.

An appealing frequency outside the permitted set cannot become the operating dial.

3 / PROPOSE & CONFIRM

A2-057 example dial

13.4151 MHz USB

Read the exact voice dial from the plan notes. Agree it with AU8 and confirm who calls first. Tune and verify actual radio readback before announcing readiness.

Centre: 13.4165 MHz. Example voice offset: −1.4 kHz. The plan supplies the dial; this is not a universal offset rule.

Why A2-057 wins here: its first three readings are F / F / F (minimum score 3), ahead of % / F / F (minimum 2). Its later X readings carry caution. A peer agreement and the normal clear-channel/RF checks are still required before a call.

SIMULATED WALKTHROUGH

03E / Keep the source beside the recommendation.

The original fixed-width report remains available in the app’s “Raw Bureau report” section and through its source link.

```
======================================================================
GRAFEX INPUT DATA
Date: 22 10 2026
T index: 60
Tx name: TX  Tx latitude: -41.14  Tx longitude: 175.04
Rx name: RX  Rx latitude: -33.87  Rx longitude: 151.18
======================================================================


SWS GRAFEX HF FREQUENCY PREDICTIONS                          ADDRESS NO. 1234
================================================================================
Circuit: TX         RX                Date:  22 October, 2026    T-index:  60
Bearings: 283 118                     Distance: 2245 km
First Mode        |--------F r e q u e n c y  (MHz)---------|       Second Mode
1F 7-11  1E 0     1   5   10    15   20    25   30    35   40    2F 21-28 2E 6
UT  OWF EMUF  ALF |...|....| ....|....| ....|....| ....|....|  OWF EMUF  ALF UT
00 17.6  0.0 11.6       ASSS XXXFFFF%%% ....                   10.4 13.8  6.8 00
01 17.9  0.0 11.7       ASSS XXXFFFF%%% ....                   10.5 13.9  6.9 01
02 17.8  0.0 11.7       ASSS XXXFFFF%%% ....                   10.5 13.7  6.9 02
03 17.3  0.0 11.4       ASSS XXXFFFF%%% ....                   10.5 13.3  6.8 03
04 16.5  0.0 11.0       ASSS XXMFFF%%%% ...                    10.3 12.5  6.6 04
05 15.3  0.0 10.3       SSSS XMFFF%%%%. ..                     10.1 11.4  6.2 05
06 15.1  0.0  9.3      ASSSM MMFFF%%%%. ..                     10.0  9.8  5.7 06
07 15.0  0.0  7.5     ASSMMM MMFF%%%%%. ...                    10.0  6.6  4.8 07
08 14.4  0.0  0.0 MMMMMMMMMM MMFF%%%%%. ...                     9.6  0.0  2.6 08
09 13.5  0.0  0.0 XMMMMMMMMM MFF%%%%... .                       9.0  1.8  0.0 09
10 12.4  0.0  0.0 XMMMMMMMMM FF%%%%...                         8.4  1.8  0.0 10
11 11.4  0.0  0.0 XMMMMMMMMM F%%%%...                          7.8  1.8  0.0 11
12 10.9  0.0  0.0 XMMMMMMMMF %%%%....                         7.5  1.8  0.0 12
13 10.5  0.0  0.0 XMMMMMMMMF %%%%...                          7.1  1.8  0.0 13
14  9.8  0.0  0.0 XMMMMMMMF% %%%...                           6.8  1.8  0.0 14
15  9.1  0.0  0.0 XMMMMMMMF% %%..                             6.2  1.8  0.0 15
16  8.4  0.0  0.0 XMMMMMMF%% ...                              5.9  1.8  0.0 16
17  7.8  0.0  0.0 XMMMMMM%%% ..                               5.7  1.8  0.0 17
18  8.1  0.0  0.0 XMMMMMMF%% ..                               5.8  1.8  0.0 18
19 10.7  0.0  6.5     ASMMFF %%%...                           6.8  0.0  4.5 19
20 13.2  0.0  8.8      ASSMM FFF%%%...                        8.6  9.0  5.5 20
21 14.8  0.0 10.0      ASSSX MFFF%%%... .                     9.5 11.0  6.0 21
22 16.2  0.0 10.8       SSSX XXFFFF%%.. ..                   10.0 12.4  6.4 22
23 17.1  0.0 11.3       ASSS XXXFFFF%%. ...                  10.4 13.2  6.7 23
UT  OWF EMUF  ALF |...|....| ....|....| ....|....| ....|....|  OWF EMUF  ALF UT
================================================================================
|.  USABLE LESS THAN 50% OF DAYS       |%  USABLE 50% TO 90% OF DAYS          |
|B  BOTH E and F MODES 90% OF DAYS     |M  MIXED FIRST AND SECOND F MODES     |
|F  FIRST F MODE ONLY     |E  E-LAYER PROPAGATION     |P  90% E and 50-90% F  |
|S  SECOND MODES ONLY     |A  HIGH ABSORPTION         |X  COMPLEX MODES       |
(c) 23-Oct-2026 BoM Space Weather Services, Sydney Australia +61-2-9213-8000
```

WORKED-EXAMPLE PROVENANCE

Trace the choice back to its inputs.

The same NZ1 → AU8 report

Prediction date: **22 October 2026**
TX: **−41.14, 175.04**
RX: **−33.87, 151.18**
T-index: **60**
Parser: `grafex-hourly-1`

The supplied example was fetched again from the public Bureau service and parsed with the app’s existing code. Its text, 24 rows and ranked channel sequence match the worked example.

What is replayed

These pages use the retained real forecast with an offline app fixture at 20Z. The candidate channels come from the supplied table. They do not establish a current channel authorization or a radio contact.

Keep forecast and evidence separate

The forecast supports a frequency proposal. Actual TX/RX recordings and reception reports establish the contact result.

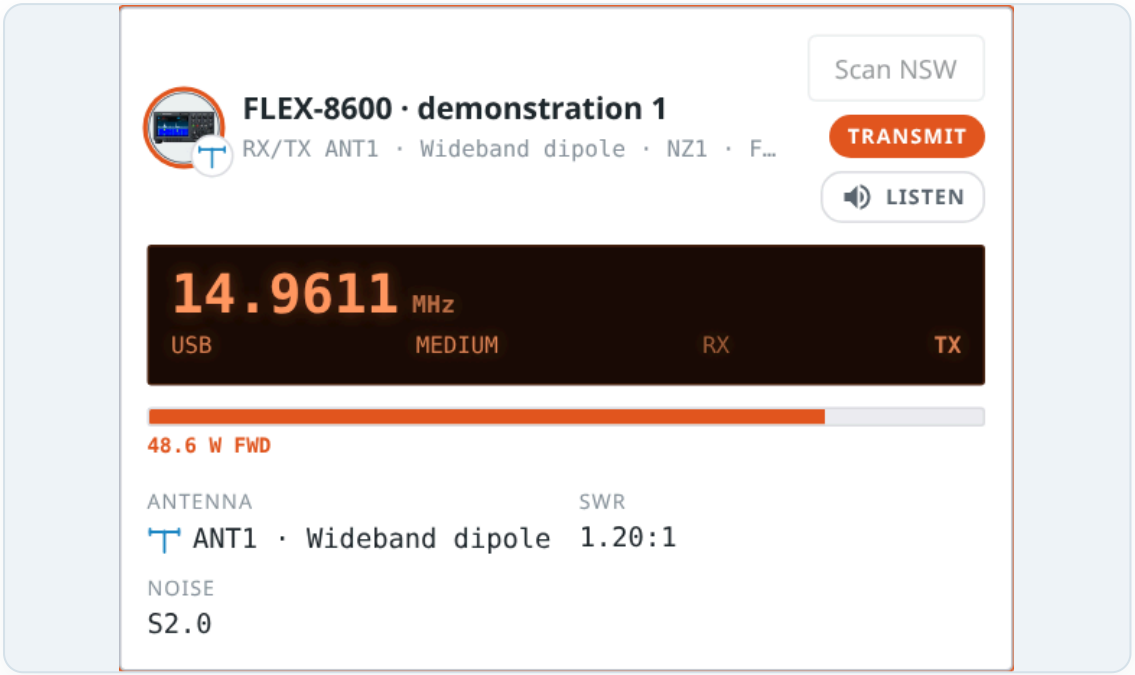
Open the exact Bureau result
Open the Bureau prediction tool

The original report, parsed JSON and reproducible offline fixtures accompany this PDF.

SIMULATED WALKTHROUGH

04 / Make one guarded voice call.

A fresh listening acknowledgement is followed by the normal preparation and clear-channel checks.



Simulated TX state in the real radio card: the agreed dial, antenna, SWR, forward power and key state are visible.

Prepare

Claim the source, verify its settings, and confirm the tuner/antenna state.

Call

Clear-channel check, then a bounded voice radio check.

Unkey

Confirm PTT released before opening the return receive window.

ONE RF PIPELINE

The call has to pass its gates.

Authorization

Approved dial, mode, call budget, station and control ownership.

Readiness

Recent peer listening reply, exact agreement and current radio readback.

RF condition

Antenna/tuner recovery, SWR, permitted power and a clear channel.

Stop conditions

EMCON, manual takeover, changed settings, duration limits or unreliable state.

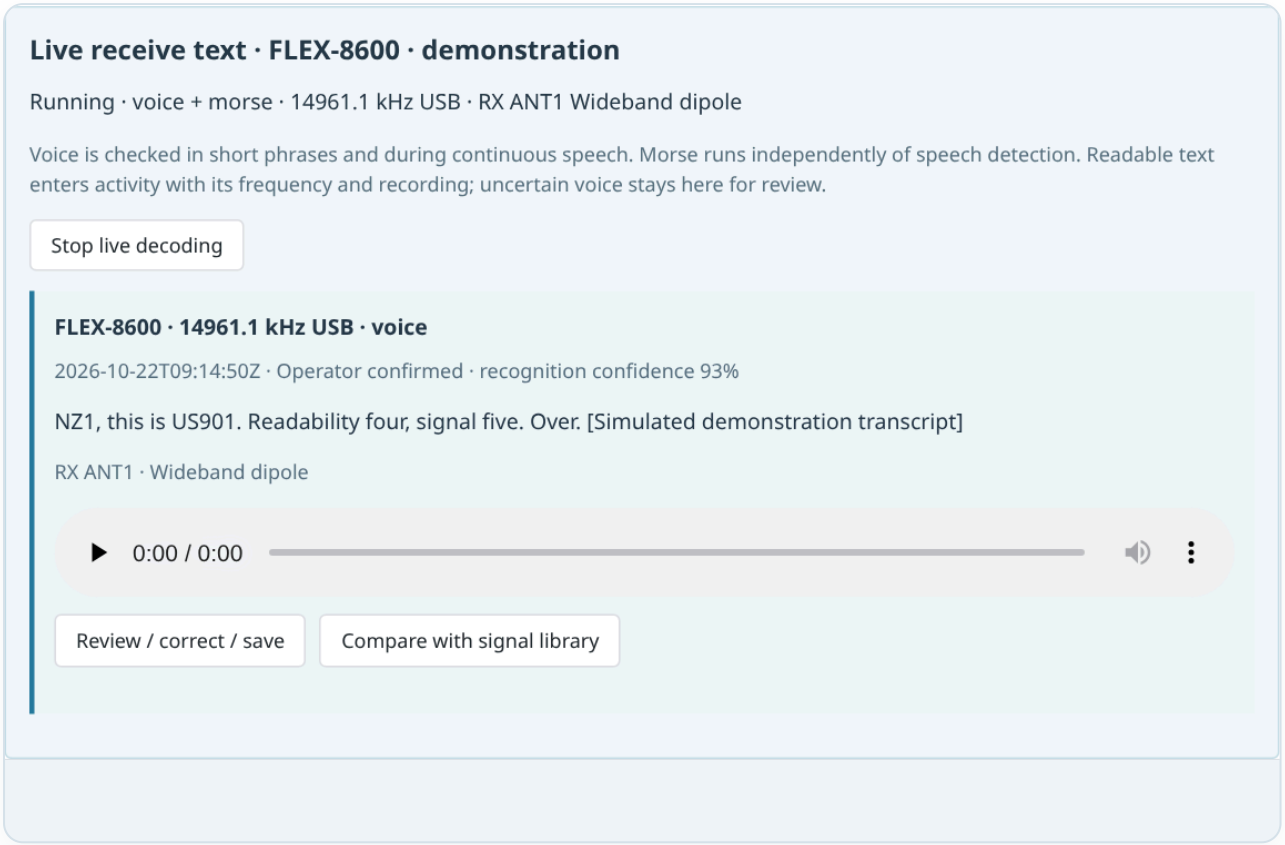
A transmission is evidence of an attempt

It becomes a successful outbound check only when the peer reports hearing NZ1. The return direction is assessed separately.

SIMULATED WALKTHROUGH

05 / Listen, decode and keep the original.

Incoming audio becomes a timed recording with its source, dial, mode and antenna context.



Actual live-receive view with an explicitly simulated US901 transcript. The app keeps decoder text and operator review distinct.

CAPTURE

Original audio

Keep the receive WAV and capture context. The original remains available for playback and download.

ASSESS

Speech and callsign

Check speech, callsign and recognition confidence. Uncertain words stay available for review.

REVIEW

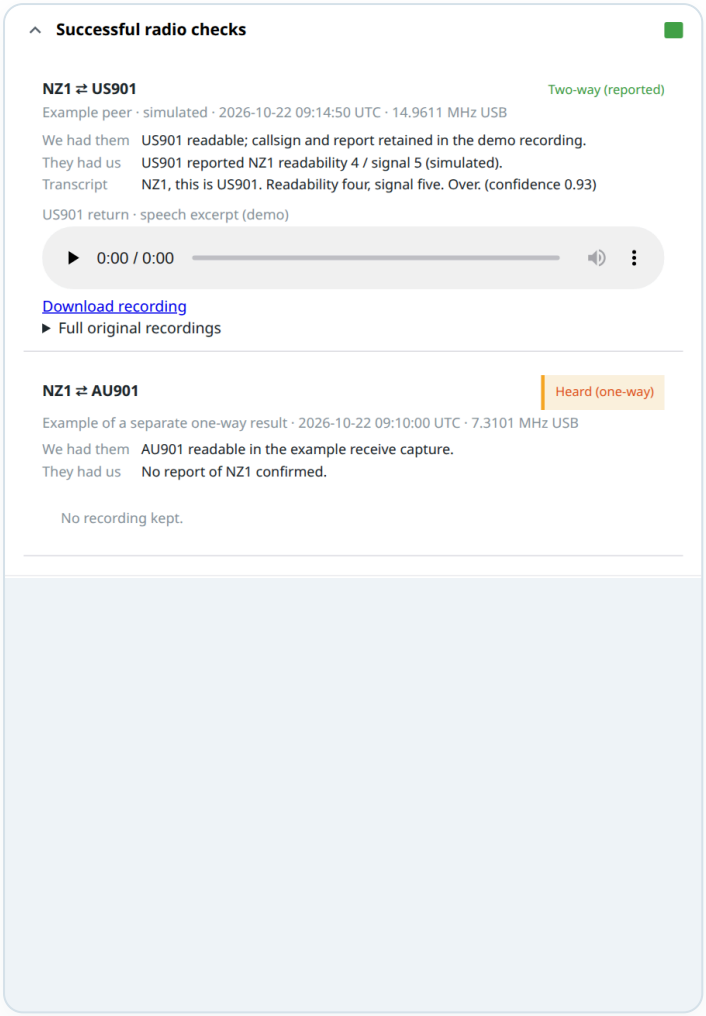
Human judgement

Listen again, correct text, confirm an observation, or keep it unresolved when the evidence is weak.

SIMULATED WALKTHROUGH

06 / Keep the two radio directions separate.

The contact log brings reports and recordings together without overstating what was confirmed.



Real contact-log component with a simulated two-way example and a separate one-way example.

EVIDENCE → RESULT

Two questions, two answers.

Did NZ1 hear the peer?

Use the recorded return, callsign and receive assessment. An operator hearing note retains its own provenance.

Did the peer hear NZ1?

Use the station's actual reception report. "Ready" and "Listening" describe readiness, not reception.

NZ1 HEARS PEER	PEER HEARS NZ1	RESULT
Yes	**Yes**	**Two-way**
Yes	No / unknown	RX only
No / unknown	Yes	TX only
Uncertain	Uncertain	**Unresolved**

An antenna-aborted transmission remains an interrupted attempt. It does not establish a clean failed propagation result.

SIMULATED WALKTHROUGH

07 / Turn confirmed evidence into a report.

A confirmed two-way result can create a draft and a visible five-minute reporting countdown.

LOG IN WEB APP: NZ1 → US901 14.9611 MHz USB · 09:13Z **3:45 to report** Copy Open Web App Mark reported

Scoring report preparation

Prepare known contact details for copying into the official Web App. The five-minute deadline runs from the contact's end, for both teams. Receive-only recordings do not confirm a two-way contact.

NZ1 → US901 · automatic draft, not submitted

3:45 to report

Phase 2.3 · demonstration

```
NZ1 → US901
Contact ended: 2026-10-22T09:13:45Z
Dial: 14.9611 MHz USB
Phase: Phase 2.3 · demonstration
Both ways operator-confirmed: yes
SIMULATED EXAMPLE ONLY.
RX: NZ1 received US901.
TX: US901 reported NZ1 readability 4 / signal 5.
Both teams report by: 2026-10-22T09:18:45Z
```

Copy contact details Open official Web App

Simulated local reporting draft. No external contact submission or official score was created for this showcase.

1 **Prepare known details**

Carry forward callsigns, contact end time, exact dial, mode and the separate reports.

2 **Keep the deadline visible**

The clock starts at the contact's end. The banner stays visible while the operator works elsewhere in the app.

3 **Record the real submission outcome**

Copy to the official Web App, or use explicitly authorized automatic submission. Preserve the returned status and receipt.

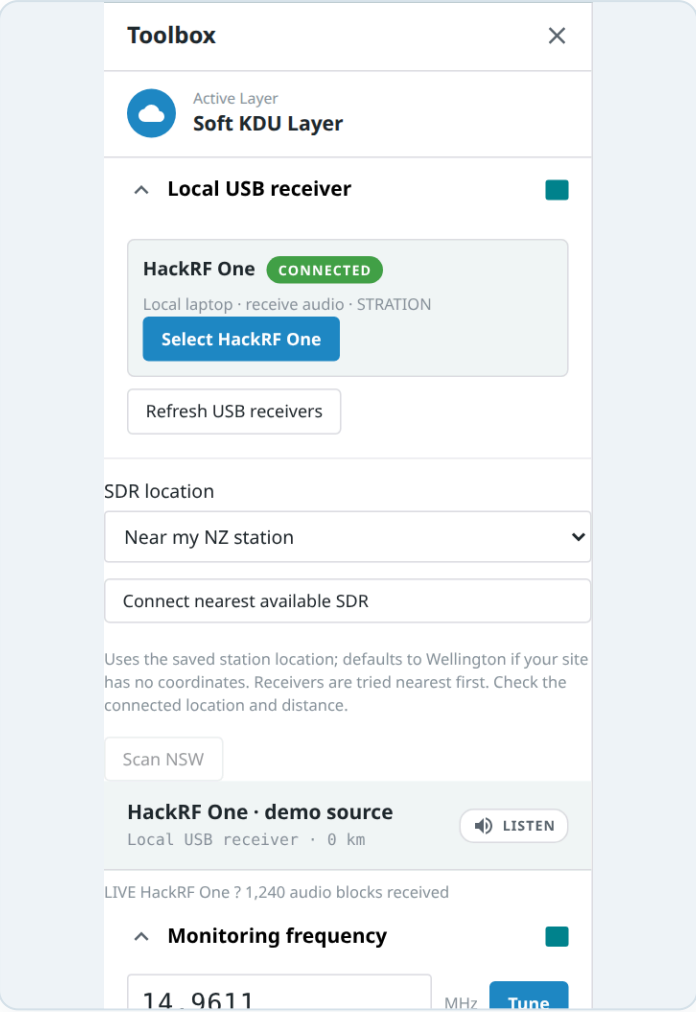
Local draft ≠ official score

"Prepared," "submitted," and the official Web App result are separate states. A receive recording alone does not create a confirmed two-way report.

SIMULATED WALKTHROUGH

Extra ears feed the same receive workflow.

The app can monitor the local radio, a local HackRF USB source or a remote KiwiSDR receiver.



Actual Soft KDU receiver controls, showing a simulated HackRF connection and audio state.

FLEX / local radio audio

The radio's receive audio enters the shared capture and speech path, alongside its telemetry and antenna context.

HackRF One / local USB

Discover and connect the USB receiver, tune its mode and dial, and select Listen. Its level is relative dBFS, not a calibrated S-meter report.

KiwiSDR / another location

Choose a public receiver near the local or distant station. Its audio describes reception at that receiver's site.

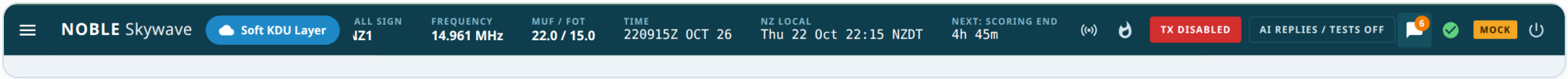
RX source → audio → recording → decode → review

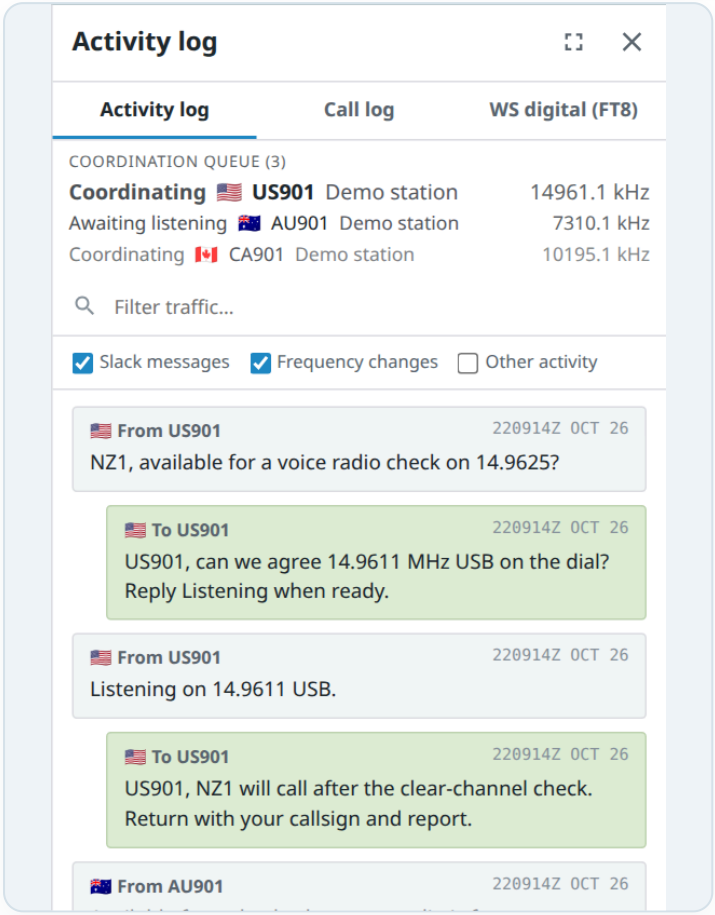
HackRF and KiwiSDR are receive-only sources in this app. A receiver may follow a radio's frequency; that does not give it transmit controls or establish what another station heard.

SIMULATED WALKTHROUGH

The operator can always see and intervene.

The activity log, queue and stop controls show the desk's current work and what still needs attention.





Actual activity log. Slack messages and frequency changes are the default filters.

1 Follow the next useful step

Calls, coordination, waiting for a listener and awaiting reports are different queue states. Eight conversations do not mean eight simultaneous transmissions.

2 Take control when needed

Manual radio changes and DM takeovers hold the affected automation. AI replies/tests and EMCON are independent controls.

3 Hand over a recoverable state

Retain actual dial, key state, active claims, remaining allowance, manual conversations, evidence and unfinished reporting work.

What a stop means

EMCON blocks RF transmission. The global automation stop and manual handoffs remain effective even when a peer is ready or a request is queued.

SIMULATED WALKTHROUGH

How the pieces connect.

A browser console brings coordination and radio evidence together through the same guarded service.

INPUTS

People and messages

Main chat, peer DMs, station roster, listening replies and reception reports.

Plans and forecasts

Approved channels, exercise phase, permitted power and path-planning information.

Radio and audio sources

FLEX telemetry and audio; HackRF USB and KiwiSDR receive streams.

APP WORKFLOW

React Soft KDU and layers

Show the same stations, controls, activity and evidence to the operator.

Guarded control service

Coordinate peers; verify readiness, claims, dial, antennas, SWR, power and stop conditions.

Evidence processing

Capture audio, decode speech or CW, support review and reconcile the two radio directions.

OUTPUTS

Station coordination and RF checks

Visible messages, bounded authorized TX, receive windows and trustworthy state changes.

Durable local evidence

Recordings, original transcripts, operator corrections, logs and contact proof packets.

Reporting and handover

Two-way drafts, deadline prompts, authorized submission results and a resumable operator state.

Request → agreement → verified radio → exchange → evidence → confirmed result → report

What this showcase demonstrates

A continuous workflow from a station asking for a check, through GRAFEX-assisted frequency agreement, to an auditable contact result. Screenshots use the built app with offline fixtures. The AU8 forecast is a verified Bureau example; demo RF exchanges and outcomes are illustrative. No live radio was keyed, no external messages were sent, and no official scores were submitted.

Prepared from the local app and its implemented control, desk, audio, receive, logging and reporting components. Supporting screenshots and the reproducible capture/build scripts are kept beside this PDF in the project's documentation folder.